

# Case Study: Behavioral Data Science

## Optimizing Product-Led Growth at BauData Chile

**Author:** Nicolás Rodríguez

**Role:** Product & Behavioral Data Analyst

### Executive Summary

This project implements a **Behavioral Data Science** framework to transition BauData from a linear sales funnel to a **Flywheel Model**. By analyzing 1,600+ events and 106 sessions via PostHog, we identified critical friction points affecting 67.92% of users and isolated a high-value "Power User" segment to drive retention and organic growth.

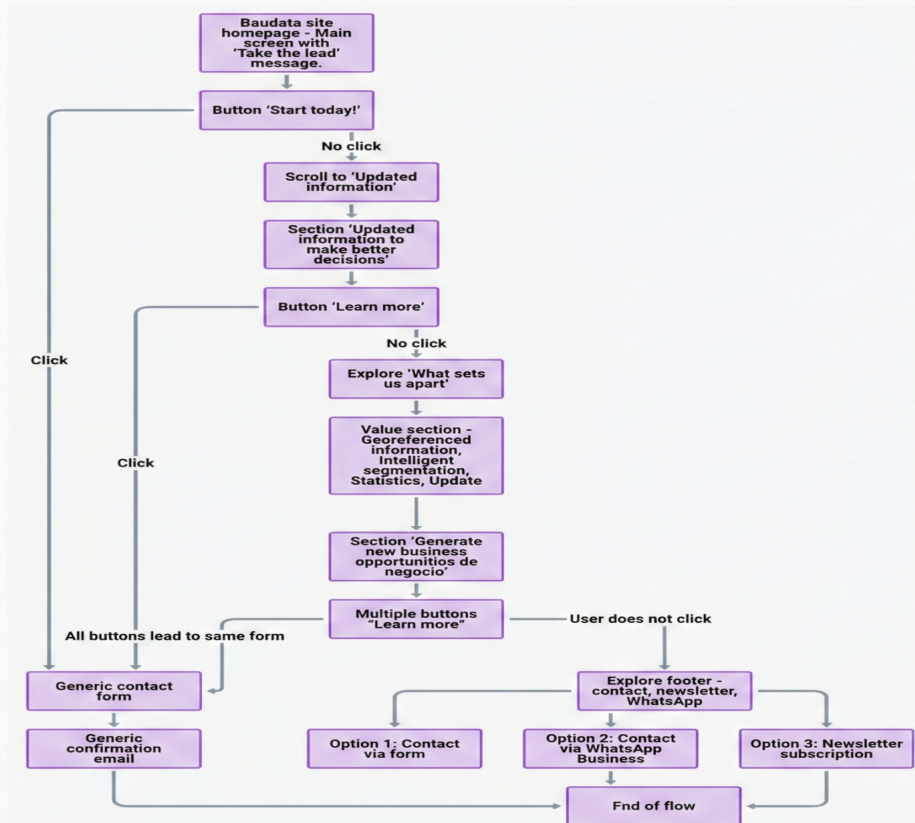
### 1. The Strategic Challenge

The construction sector in Chile traditionally relies on linear sales processes. BauData, a PropTech spin-off, needed to evolve its business model. The objective was to diagnose adoption barriers and design an ecosystem where user satisfaction and recurring data usage become the primary drivers for growth.

### 2. Behavioral Audit & Attraction Phase

Prior to technical intervention, a digital psychology audit identified critical cognitive biases and misalignments in the attraction phase:

- **Choice Overload:** The interface presented excessive data density, overwhelming the user's cognitive load and causing preemptive abandonment.
- **Anchoring Bias:** A lack of clear narrative regarding data value prevented users from "anchoring" service benefits against the learning curve.
- **2.5 Semantic Intent Mapping (SEO & Attraction):** Beyond site interactions, we analyzed the "Search Intent" that brings users to the platform. By clustering keywords (e.g., "Social Housing" vs. "Private Infrastructure"), we identified that the **Attract** phase often brought users with highly specific needs who faced immediate friction if the landing page didn't semantically match their search query. This misalignment contributes to the high bounce rates seen in the broader dataset.



### 3. Technical Approach & Pipeline

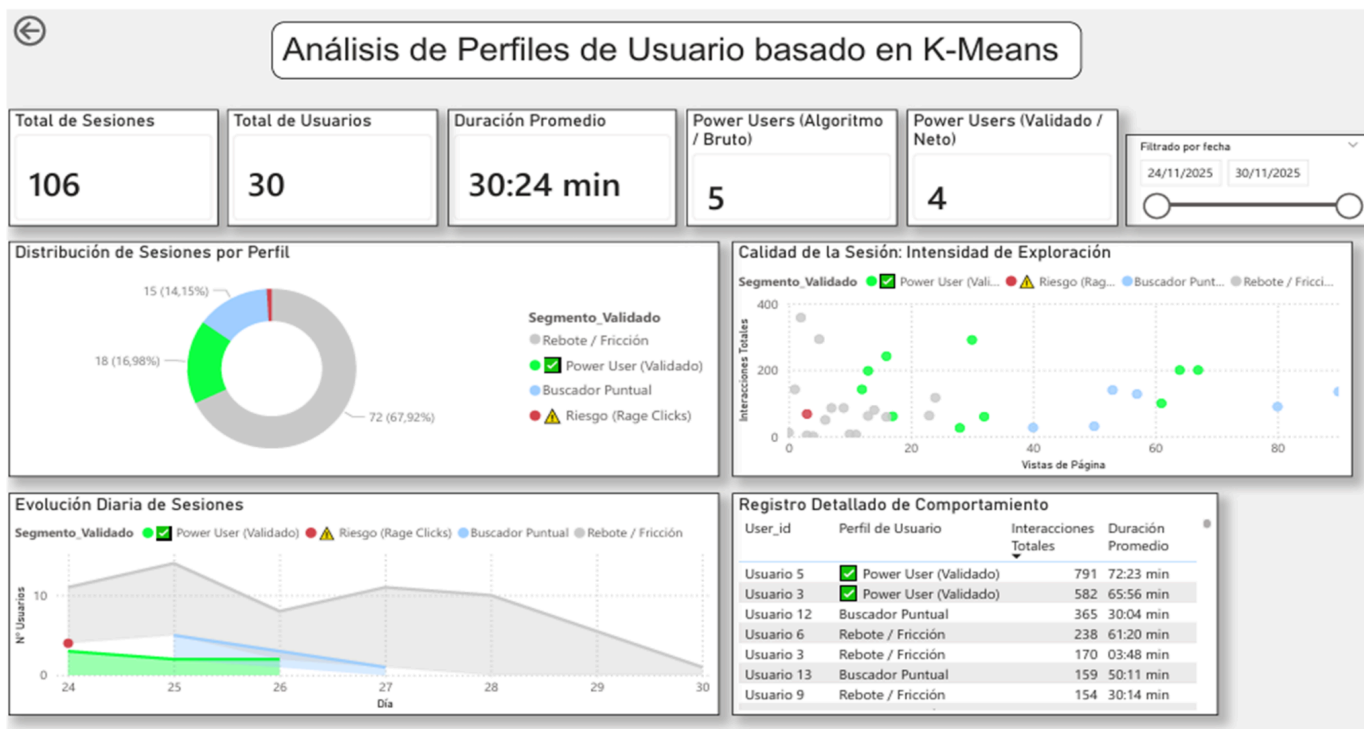
A real-time telemetry pipeline was implemented to validate behavioral hypotheses.

- **Data Ingestion:** 1,623 event records and 106 unique sessions via PostHog.
- **Feature Engineering:** Developed custom metrics such as `real_duration` (active time) and `marker_select_count` to capture true user intent.
- **Modeling:** Executed a **K-Means Clustering (k=3)** model following `StandardScaler` normalization to accurately segment interaction profiles.

### 4. Strategic Findings: User Segmentation

The model revealed three archetypes that redefined the product roadmap:

1. **Power Users (14.15%):** Users with an average active duration of **409 seconds**. They are the engine of the Flywheel.
2. **Targeted Searchers (16.98%):** High-efficiency users with specific search intent but low exploration.
3. **Bounce & Friction (67.92%):** The most critical finding; users who abandoned the flow due to onboarding barriers.



## 5. Impact Proposals & Ethical Governance

- **UI Redesign:** Simplification of the onboarding process to mitigate cognitive overload.
- **Recommendation Engine:** Logic designed to offer tailored projects based on historical behavior.
- **Human-in-the-loop:** A governance framework to ensure transparency in territorial data management across **Chile's 340 municipalities**.

*This case study demonstrates the intersection of Data Science and Behavioral Science in the PropTech market.*